

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `c:\Users\sarah\OneDrive\Documents\GitHub\hw5-sarah\_nathan\_hw5`

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
# Component table (mass share, ash %, MRF recycle %)
comp = [
    (0.15, 0.08, 0.00),    # Food waste
    (0.40, 0.07, 0.55),    # Paper and cardboard
    (0.05, 0.05, 0.15),    # Plastics
    (0.03, 0.10, 0.10),    # Textiles
    (0.02, 0.15, 0.00),    # Rubber, leather
    (0.05, 0.02, 0.30),    # Wood
    (0.18, 0.02, 0.40),    # Yard wastes
    (0.04, 1.00, 0.60),    # Glass
    (0.02, 1.00, 0.75),    # Ferrous
    (0.02, 1.00, 0.80),    # Aluminum
    (0.01, 1.00, 0.50),    # Other metals
    (0.03, 0.70, 0.00)     # Misc
```

```

]

# Calculate fractions
recycle = sum(x[1] * x[3] for x in values(comp))
ash = sum(x[1] * x[2] for x in values(comp))

println("Recycling fraction: ", round(recycle, digits=3))
println("Ash fraction: ", round(ash, digits=3))

```

Recycling fraction: 0.378
 Ash fraction: 0.164

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Decision variables

W_{ij} Waste transported from source i to disposal j (mg/day)

R_{kj} Residual waste transported from disposal k to disposal j (mg/day)

Y_j Operational status of disposal j (binary); 0 if facility not operated and 1 if facility operated

Additional variables

i = cities 1,2,3

j = LF, MRF, WTE

d_{ij} Distance between city i and facility j

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Objective: minimize Total Cost

= Fixed Costs + Transportation Costs + Disposal Costs (tipping/recycling)

Fixed cost = $2000Y_{LF} + 1500Y_{MRF} + 2500Y_{WTE}$

Fixed cost for each facility multiplied by operational status

Transportation cost = $\$ 1.5 \sum_{i=1}^3 \sum_{j=LF,MRF,WTE} d_{ij} W_{ij}$

Multiply fixed transportation cost by sum of distance between cities and facilities and waste sent from each city to each facility

Tipping cost = $\$ 50(\sum_i W_{i,LF} + R_{MRF,LF} + R_{WTE,LF}) + 40(\sum_i W_{i,MRF}) + 60(\sum_i W_{i,WTE} + R_{MRF,WTE})$

Multiply tipping cost for each facility by waste input into facility and residual waste delivered by other facilities

Recycling cost = $40(0.4W_{i,MRF})$

40% recycling rate and \$40/mg

Objective function

minimize $2000Y_{LF} + 1500Y_{MRF} + 2500Y_{WTE} + 1.5 \sum_{i=1}^3 \sum_{j=LF,MRF,WTE} d_{ij} W_{ij} + 50(\sum_i W_{i,LF} + R_{MRF,LF} + R_{WTE,LF}) + 40(\sum_i W_{i,MRF}) + 60(\sum_i W_{i,WTE} + R_{MRF,WTE})$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

```
# Data

facilities = [:LF, :MRF, :WTE]

# City waste generation (mg/d)
cities = 1:3
waste = Dict{1 => 100.0, 2 => 90.0, 3 => 120.0}
resid_source = [:MRF, :WTE]

# Distances from facilities to cities (km)
dist_city = Dict{
    :LF => Dict{1 => 5.0, 2 => 15.0, 3 => 13.0},
    :MRF => Dict{1 => 30.0, 2 => 25.0, 3 => 45.0},
    :WTE => Dict{1 => 15.0, 2 => 10.0, 3 => 20.0}
}
```

```

# Distances between facilities (km)
dist_fac = Dict(
    (:MRF, :LF) => 32.0,
    (:MRF, :WTE) => 15.0,
    (:WTE, :LF) => 18.0
)

# Capacities (Mg/day)
cap = Dict(:LF => 200.0, :MRF => 350.0, :WTE => 210.0)

# Fixed costs ($/day)
F = Dict(:LF => 2000.0, :MRF => 1500.0, :WTE => 2500.0)

# Tipping costs ($/mg)
tipping = Dict(:LF => 50.0, :MRF => 7.0, :WTE => 60.0)

transport = 1.5      # Transportation costs ($/mg-km)
recycle = 40.0       # Recycling costs ($/mg recycled)

MRF_recycle = 0.4    # MRF recycling rate
ash_NR = 0.16        # Ash fraction of non-recycled waste
ash_recycle = 0.14   # Ash fraction of recycled waste

# Composition table (mass share, ash %, MRF recycle %)
comp = Dict(
    :Food           => (0.15, 0.08, 0.00),
    :PaperCardboard => (0.40, 0.07, 0.55),
    :Plastics       => (0.05, 0.05, 0.15),
    :Textiles       => (0.03, 0.10, 0.10),
    :RubberLeather  => (0.02, 0.15, 0.00),
    :Wood           => (0.05, 0.02, 0.30),
    :Yard           => (0.18, 0.02, 0.40),
    :Glass          => (0.04, 1.00, 0.60),
    :Ferrous        => (0.02, 1.00, 0.75),
    :Aluminum       => (0.02, 1.00, 0.80),
    :OtherMetal     => (0.01, 1.00, 0.50),
    :Misc           => (0.03, 0.70, 0.00)
)

total_cost = Model(HiGHS.Optimizer)

@variable(total_cost, W[i in cities, j in facilities] >= 0)    # W_ij

```

```

@variable(total_cost, R[k in resid_source, j in facilities] >= 0)      # R_kj
@variable(total_cost, Y[j in facilities], Bin)                        # Y_j

@constraint(total_cost, [i in cities],                                # All waste is delivered somewhere
    sum(W[i, j] for j in facilities) == waste[i]
)

@objective(total_cost, Min,
    # Fixed cost
    2000 * Y[:LF] + 1500 * Y[:MRF] + 2500 * Y[:WTE] +

    # Transport cost
    1.5 * sum(dist_city[j][i] * W[i, j] for i in cities, j in facilities) +

    # Tipping and recycling
    50 * (sum(W[i, :LF] for i in cities) + R[:MRF, :LF] + R[:WTE, :LF]) + 40 * sum(W[i, :MRF, :LF] for i in cities) +
    60 * (sum(W[i, :WTE] for i in cities) + R[:MRF, :WTE])
)

optimize!(total_cost)
println("\nTotal Cost: ", objective_value(total_cost))

```

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms
MIP has 3 rows; 18 cols; 9 nonzeros; 3 integer variables (3 binary)

Coefficient ranges:

```

Matrix  [1e+00, 1e+00]
Cost    [5e+01, 2e+03]
Bound   [1e+00, 1e+00]
RHS     [9e+01, 1e+02]

```

Presolving model

3 rows, 3 cols, 3 nonzeros 0s

0 rows, 0 cols, 0 nonzeros 0s

Presolve reductions: rows 0(-3); columns 0(-18); nonzeros 0(-9) - Reduced to empty

Presolve: Optimal

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristic;

I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Randomized rounding;

S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y => HiGHS solution;

Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper; z => Trivial upper bound;

	Nodes		B&B Tree		Objective Bounds		Dynamic Cuts		
Src	Proc.	InQueue	Leaves	Expl.	BestBound	BestSol	Gap	Cuts	In

0	0	0	0.00%	20615	20615	0.00%	0
---	---	---	-------	-------	-------	-------	---

Solving report

Status	Optimal
Primal bound	20615
Dual bound	20615
Gap	0% (tolerance: 0.01%)
P-D integral	0
Solution status	feasible
	20615 (objective)
	0 (bound viol.)
	0 (int. viol.)
	0 (row viol.)
Timing	0.00
Max sub-MIP depth	0
Nodes	0
Repair LPs	0
LP iterations	0

Total Cost: 20615.0

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in ‘data/generators.csv.’ In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

```
G = ["Biomass","Hydro","Geo","NG_CCGT","NG_CT","Wind","Solar"] #G is the different generation

Pmin = Dict("Biomass"=>0,"Hydro"=>0,"Geo"=>0,"NG_CCGT"=>100,"NG_CT"=>100,"Wind"=>0,"Solar"=>0)
Pmax = Dict("Biomass"=>100,"Hydro"=>500,"Geo"=>400,"NG_CCGT"=>250,"NG_CT"=>250,"Wind"=>300,"Solar"=>0)
Cost = Dict("Biomass"=>5,"Hydro"=>0,"Geo"=>0,"NG_CCGT"=>23,"NG_CT"=>38,"Wind"=>0,"Solar"=>0)
Ramp = Dict("Biomass"=>100,"Hydro"=>500,"Geo"=>400,"NG_CCGT"=>100,"NG_CT"=>200,"Wind"=>300,"Solar"=>0)

d1 = 1100.0 #demand in period 1

#different scenarios for period 2 from the tree, including (name, demand, solar_cf, wind_cf,
scenarios = [
    ("S1", 1200.0, 0.95, 0.40, 0.525),
    ("S2", 1200.0, 0.75, 0.50, 0.225),
    ("S3", 1500.0, 0.95, 0.40, 0.175),
    ("S4", 1500.0, 0.75, 0.50, 0.075)
]

#renewable output (fixed)
solar1 = Pmax["Solar"] * 0.90 #500 * .9 = 450
wind1 = Pmax["Wind"] * 0.45 #300 * .45 = 135

m = Model(HiGHS.Optimizer) #write the optimization model

#p1(g) = generation in period 1 MWh
@variable(m, p1[g in G], lower_bound=Pmin[g], upper_bound=Pmax[g])

#fix renewables p1 so they are not decision variables
fix(p1["Solar"], solar1; force=true) #help from stackoverflow, still unsure what this does
fix(p1["Wind"], wind1; force=true)

#p2(g,s)
@variable(m, p2[g in G, s in 1:4], lower_bound=Pmin[g], upper_bound=Pmax[g])

for s in 1:length(scenarios)
    scenario = scenarios[s]

    solar_cf = scenario[3] #third
    wind_cf = scenario[4] #fourth
```

```

    fix(p2["Solar", s], Pmax["Solar"] * solar_cf; force=true)
    fix(p2["Wind", s], Pmax["Wind"] * wind_cf; force=true)
end

#p1
@constraint(m, sum(p1[g] for g in G) == d1)

#p2 (one per scenario)
for s in 1:length(scenarios)
    d2 = scenarios[s][2] #second
    @constraint(m, sum(p2[g, s] for g in G) == d2)
end

#ramping constraints
for g in G
    if g != "Solar" && g != "Wind"
        RU = Ramp[g]
        for s in 1:length(scenarios)
            @constraint(m, p2[g,s] - p1[g] <= RU)
            @constraint(m, p1[g] - p2[g,s] <= RU)
        end
    end
end

end

#objective
@objective(m, Min,
    sum(Cost[g] * p1[g] for g in G) +
    sum(
        scenarios[s][5] * Cost[g] * p2[g,s]
        for s in 1:length(scenarios), g in G
    )
)

#optimize
optimize!(m)

#start printing the results
println("Expected cost = ", objective_value(m))

println("\np1:")
for g in G
    println(g, ": ", value(p1[g]))
end

println("\np2 by scenario:")
for s in 1:length(scenarios)

```

```

    name = scenarios[s][1]
    println(name)
    for g in G
        println("  ", g, ": ", value(p2[g,s]))
    end
end
end

```

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms
 LP has 45 rows; 35 cols; 115 nonzeros

Coefficient ranges:

```

Matrix [1e+00, 1e+00]
Cost   [4e-01, 4e+01]
Bound  [1e+02, 5e+02]
RHS    [1e+02, 2e+03]

```

Presolving model

13 rows, 19 cols, 35 nonzeros 0s

6 rows, 11 cols, 16 nonzeros 0s

4 rows, 5 cols, 8 nonzeros 0s

Presolve reductions: rows 4(-41); columns 5(-30); nonzeros 8(-107)

Solving the presolved LP

Performed postsolve

Solving the original LP from the solution after postsolve

```

Model status      : Optimal
Objective value    : 1.2200000000e+04
P-D objective error : 0.0000000000e+00
HiGHS run time     : 0.01
Expected cost = 12200.0

```

p1:

Biomass: 0.0

Hydro: 315.0

Geo: 0.0

NG_CCGT: 100.0

NG_CT: 100.0

Wind: 135.0

Solar: 450.0

p2 by scenario:

S1

Biomass: 0.0

Hydro: 405.0
Geo: 0.0
NG_CCGT: 100.0
NG_CT: 100.0
Wind: 120.0
Solar: 475.0

S2

Biomass: 0.0
Hydro: 475.0
Geo: 0.0
NG_CCGT: 100.0
NG_CT: 100.0
Wind: 150.0
Solar: 375.0

S3

Biomass: 0.0
Hydro: 305.0
Geo: 400.0
NG_CCGT: 100.0
NG_CT: 100.0
Wind: 120.0
Solar: 475.0

S4

Biomass: 0.0
Hydro: 375.0
Geo: 400.0
NG_CCGT: 100.0
NG_CT: 100.0
Wind: 150.0
Solar: 375.0

References

List any external references consulted, including classmates.